

Robert cross operation:

- Introduced in 1963. It is a 2×2 kernel or mask,
 - This is a forward difference approximation.
 - This operator compute a derivative along x -derivative
- diff are $g_x = f(x, y) - f(x+1, y+1)$, $g_y = f(x+1, y) - f(x, y+1)$

2 Roberts cross Mask are

$$G_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$G_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

- It uses 2×2 operator only.

convolution process:

convulsion is a process of sliding a small kernel or mask at every location.

- place kernel on img
- Multiply corresponding element
- Add all product
- store the result.
- Move kernel to next position
- this process is repeated, until entire img is processed.

step: 1 original image $f(x, y) = \begin{bmatrix} 20 & 25 & 35 & 40 \\ 30 & 40 & 60 & 65 \\ 45 & 70 & 90 & 95 \\ 55 & 80 & 110 & 120 \end{bmatrix}$

step: 2 Roberts mask

horizontal mask $G_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

vertical mask $G_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

step: 3: place a mask

the first 2×2 window

high boost filtering
 consider a following 1D pixel value 20, 25, 30, 150. calculate a derivative.

$$f(x+1, y) =$$

∴ Edge is located blur 30 to 150 : diff 120. (largest derivative)

$$20, 25, 30, 150$$

$$25 - 20 = 5$$

$$30 - 25 = 5$$

$$150 - 30 = 120 \Rightarrow \text{largest derivative.}$$

Q. $G_x = 80$, $G_y = 60$. calculate gradient magnitude and direction

$$\text{Gradient magnit (G)} = \sqrt{G_x^2 + G_y^2}$$

$$= \sqrt{6400 + 3600}$$

$$= \sqrt{10000}$$

$$G = 100$$

$$\text{Gradient direction } \theta = \tan^{-1} \left(\frac{G_y}{G_x} \right)$$

$$= \tan^{-1} \left(\frac{60}{80} \right)$$

$$= \tan^{-1} \left(\frac{3}{4} \right)$$

$$\begin{bmatrix} 20 & 25 \\ 30 & 40 \end{bmatrix}$$

compute horizontal gradient

$$y = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

multiply corresponding elements

$$G_x = (1 \times 20) + 0 + 0 + (-1 \times 40)$$

$$= 20 - 40$$

$$G_x = -20$$

$$G_y = 0 + 25 - 30 + 0$$

$$= -5$$

Gradient magnitude;

$$G = \sqrt{(20)^2 + (-5)^2}$$

$$= \sqrt{425}$$

$$G_1 = 20.62$$

2nd window;

$$G_x = 205 - 40$$

$$= -135$$

$$G_y = 0 - 5$$

$$G_2 = 15.81$$

3rd window;

$$G_x = \begin{bmatrix} -30 & 20 & 20 & 0 \\ 20 & 10 & 0 & 0 \\ -20 & 10 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$G_y = 36.05$$

4th window

$$G_x = -10$$

$$G_y = 15$$

$$G_4 = 40.07$$

5th window;

$$G_x = -50$$

$$G_y = -10$$

$$G_5 = 50.99$$

6th window;

$$G_x = -35$$

$$G_y = 25$$

$$G_6 = 43.01$$

7th window;

$$G_x = -35$$

$$G_y = 15$$

$$G_7 = 38.07$$

8th window;

$$G_x = 40$$

$$G_y = 10$$

$$G_8 = 41.23$$

9th window;

$$G_x = 30$$

$$G_y = 15$$

$$G_9 = 33.54$$

20.62	15.81	36.05
40.07	50.99	43.01
38.07	41.23	33.54

Edge → highest intensity value.

$$G_{11} = |G_{12}| + |G_{18}|$$

$$G_{11} = 25$$

$$G_{15} = 60$$

$$G_{19} = 45$$

$$G_{12} = 40$$

$$G_{16} = 50$$

$$G_{13} = 50$$

$$G_{17} = 50$$

$$G_{14} = 45$$

$$G_{18} = 50$$

Prüf operator;

comparison with edge detection

Robert operator mask size is 2×2 . Best for

diagonal edges.

This operator is also used for edge detection. It uses

3×3 mask or kernel. G_{1x} = vertical, G_{1y} = horizontal. Edge strength

$$\text{is } \sqrt{G_{1x}^2 + G_{1y}^2}$$

$$\text{Horizontal } G_{1x} = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\text{vertical } G_{1y} = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 \end{bmatrix}$$

2.

20	30	40
40	60	80
70	90	120

sol.

$$\text{Horizontal mask} = -20 - 40 - 70 + 40 + 80 + 120$$

$$= -130 + 240$$

$$= 110$$

$$\text{vertical mask} = -20 - 30 - 40 + 70 + 90 + 120$$

$$= -90 + 280$$

$$= 190$$

$$G = \sqrt{G_x^2 + G_y^2}$$

$$= \sqrt{(110)^2 + (190)^2}$$

$$= 219.54$$

$G_x = 110$ indicates the pixel intensity increases from left to right.

$G_y = 190$ indicates the pixel intensity increases from top to bottom.

Sobel operator:

It is one of most popular, used for edge detection and in sharpening. 3×3 matrix/kernel.

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Q. $img = \begin{bmatrix} 20 & 20 & 120 \\ 20 & 20 & 120 \\ 20 & 20 & 120 \end{bmatrix}$

Sol:

Horizontal mask = $-20 - 40 - 20 + 240 + 240$

$$= 480 - 80$$

$G_x = 400 \Rightarrow$ strong vertical edge

vertical mask = $-20 - 40 - 120 + 20 + 40 - 120$

$G_y = 0 \Rightarrow$ There is no vertical intensity change

$$G = \sqrt{(400)^2 + 0^2}$$

$$= 400 \Rightarrow \text{Edge value}$$

Laplacian operators (2nd order)

commonly used Laplacian mask:

$$L = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Q.

$$\text{img} = \begin{bmatrix} 10 & 20 & 30 & 40 \\ 20 & 40 & 60 & 80 \\ 40 & 60 & 80 & 100 \\ 60 & 80 & 100 & 120 \end{bmatrix}$$

sol:

$$\text{Window 1: } \begin{bmatrix} 10 & 20 & 30 \\ 20 & 40 & 60 \\ 40 & 60 & 80 \end{bmatrix}$$

$$\text{Laplacian mask} = 20 - 20 + 160 - 60 - 60$$

$$= 160 - 160$$

$$= 0$$

Window 2:

$$\begin{bmatrix} 20 & 30 & 40 \\ 40 & 60 & 80 \\ 60 & 80 & 100 \end{bmatrix}$$

$$\text{Laplacian mask} = -30 - 40 + 240 - 80 - 80$$

$$= 240 - 150 - 80$$

$$= 10$$

Window 3:

$$\begin{bmatrix} 40 & 60 & 80 \\ 60 & 80 & 100 \end{bmatrix}$$

$$\text{Laplacian} = -40 - 40 + 240 - 80 - 80$$

$$= 0$$

Window 4:

$$\begin{bmatrix} 40 & 60 & 80 \\ 60 & 80 & 100 \\ 80 & 100 & 120 \end{bmatrix}$$

Laplacian mask = $60 - 60 + 320 - 100 - 100$
 $= 0$

Comparison:

operator

Mask size

Best for

speed

noise sensitive
very

Roberts

2x2

Diagonal edge

High

Prewitt

3x3

Horizontal & vertical edge

medium

fast

Sobel

3x3

Strong edge detect.

Lower

fast

Canny

Multi-stage

Accurate edge detect

Very Low

slow

2/1/1

20/7/2020

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$08 - 02 - 015 + 01 - 05 = 10$

$08 - 021 - 015 = 10$

$01 =$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$